UNIVERSITY OF GHANA
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MSc Computer Science

Feature-based Image Matching and Automatic Panorama Construction

Project report

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A classical OpenCV pipeline (detect, describe, match, RANSAC, warp, stitch) with an interactive walkthrough. The full two-column technical paper lives on the live app's Project report tab and is not this document.

1. Where to find the work

- Live application: https://feature-based-panorama.streamlit.app/
- Source (public): https://github.com/qbentil/feature-based-panorama
- Full technical paper (CVPR LaTeX): report/feature_based_panorama.pdf in the repository, also embedded on the app's Project report tab.

This file is the short project packet: method, source excerpts, experiment figures, and screenshots of the running app. It is not the CVPR paper shown in the application.

The repository contains the pipeline under src/, the Streamlit walkthrough under app_pages/, Wikimedia Commons campus photographs (Balme Library, Dance Department, Night Market, Commonwealth Hall) with licences in data/ATTRIBUTION.md, and experiment tables in report/results/. The implementation does not call cv2.Stitcher, YOLO, or any pretrained recogniser.

2. Introduction

A panorama is a single wide image assembled from overlapping photographs of the same scene. Consumer cameras hide the geometry: they find the same landmarks in neighbouring frames, discard inconsistent correspondences, and warp one frame onto another. Those steps are standard computer vision—feature detection, description, matching, robust homography estimation, and blending—rather than a learned recogniser.

This project implements that chain explicitly so every decision is inspectable:

1. Detect distinctive keypoints and compute descriptors with SIFT and with ORB.
2. Match descriptors between overlapping pairs and visualise correspondences before and after RANSAC.
3. Estimate a homography, warp images into one coordinate system, and stitch a panorama from at least three views.
4. Compare the detectors on match count, inlier ratio, reprojection error, overlap SSIM, and time.
5. Measure how rotation, scale, illumination, and viewpoint change affect the inlier ratio.

The shared geometry lives in src/. Both the Streamlit app and scripts/run_experiments.py call the same match_pair and stitch_images entry points; the UI never reimplements the homography.

The pipeline is:

overlapping photos → prepare → detect and describe → match → RANSAC → homography → warp to the middle photo → feather blend → panorama.

3. Implementation

Each image is resized so the long edge is 1200 px, lightly Gaussian-smoothed, and optionally CLAHE-equalised on the grayscale copy used for detection. Colour is kept for warping and blending. Keypoints are capped at 2000 per image so SIFT and ORB are compared under the same budget.

3.1. Detect and describe (src/features.py)

SIFT builds a gradient-histogram descriptor designed to be stable under scale and rotation. ORB combines FAST corners with a rotated BRIEF binary descriptor; matching then uses Hamming distance.

```
def create_detector(name, n_features=2000):
    key = name.strip().upper()
    if key == "SIFT":
        return cv2.SIFT_create(nfeatures=n_features)
    if key == "ORB":
        return cv2.ORB_create(nfeatures=n_features)
```

Listing 1: Detector factory.

3.2. Match (src/matching.py)

Brute-force kNN (k=2) followed by Lowe's ratio test at 0.75. SIFT uses L_2; ORB uses Hamming.

```
knn = matcher.knnMatch(descriptors_a, descriptors_b, k=2)
good = []
for pair in knn:
    if len(pair) < 2:
        continue
    best, second = pair
    if best.distance < ratio * second.distance:
        good.append(best)
```

Listing 2: Lowe ratio test on kNN pairs.

3.3. RANSAC homography (src/homography.py)

cv2.findHomography with a 5 px reprojection threshold. Consecutive pair maps are composed into the middle image's frame so the canvas does not drift to one end of the sequence.

```
H, mask = cv2.findHomography(
    pts_src.reshape(-1, 1, 2),
    pts_dst.reshape(-1, 1, 2),
    method=cv2.RANSAC,
    ransacReprojThreshold=ransac_thresh,
)
```

Listing 3: Robust homography.

3.4. Warp and stitch (src/stitch.py)

Each view is warpPerspective-mapped onto a canvas large enough for all warped corners. Overlap is feather-blended with a distance-transform weight so seams do not appear as hard cuts.

```
dist = cv2.distanceTransform(binary, cv2.DIST_L2, 5)
w = dist.astype(np.float32)
acc += img.astype(np.float32) * w[..., None]
weight += w
blended = acc / np.maximum(weight[..., None], 1e-6)
```

Listing 4: Distance-transform feather blend.

Quality metrics reported for every pair: initial matches, RANSAC inliers, inlier ratio, mean reprojection error, and overlap SSIM.

4. Interactive application

Six tabs: How it works, Build a panorama, Compare detectors, Robustness lab, Saved results, and Project report. The last of those embeds the full CVPR paper; the screenshots below are the walkthrough a marker can click through on the live URL.

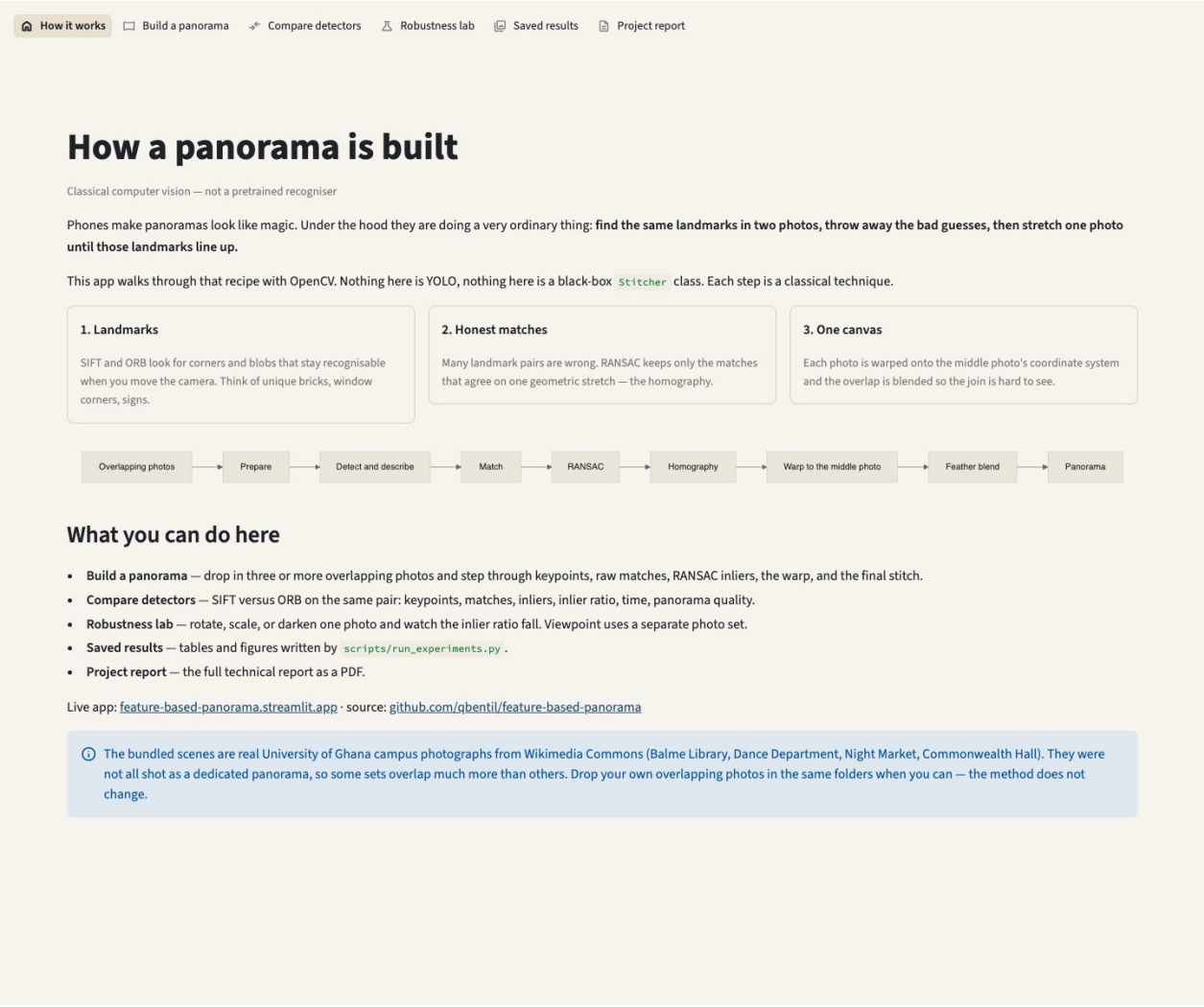


Figure 1: How it works: pipeline explanation (detect → match → RANSAC → warp → blend).

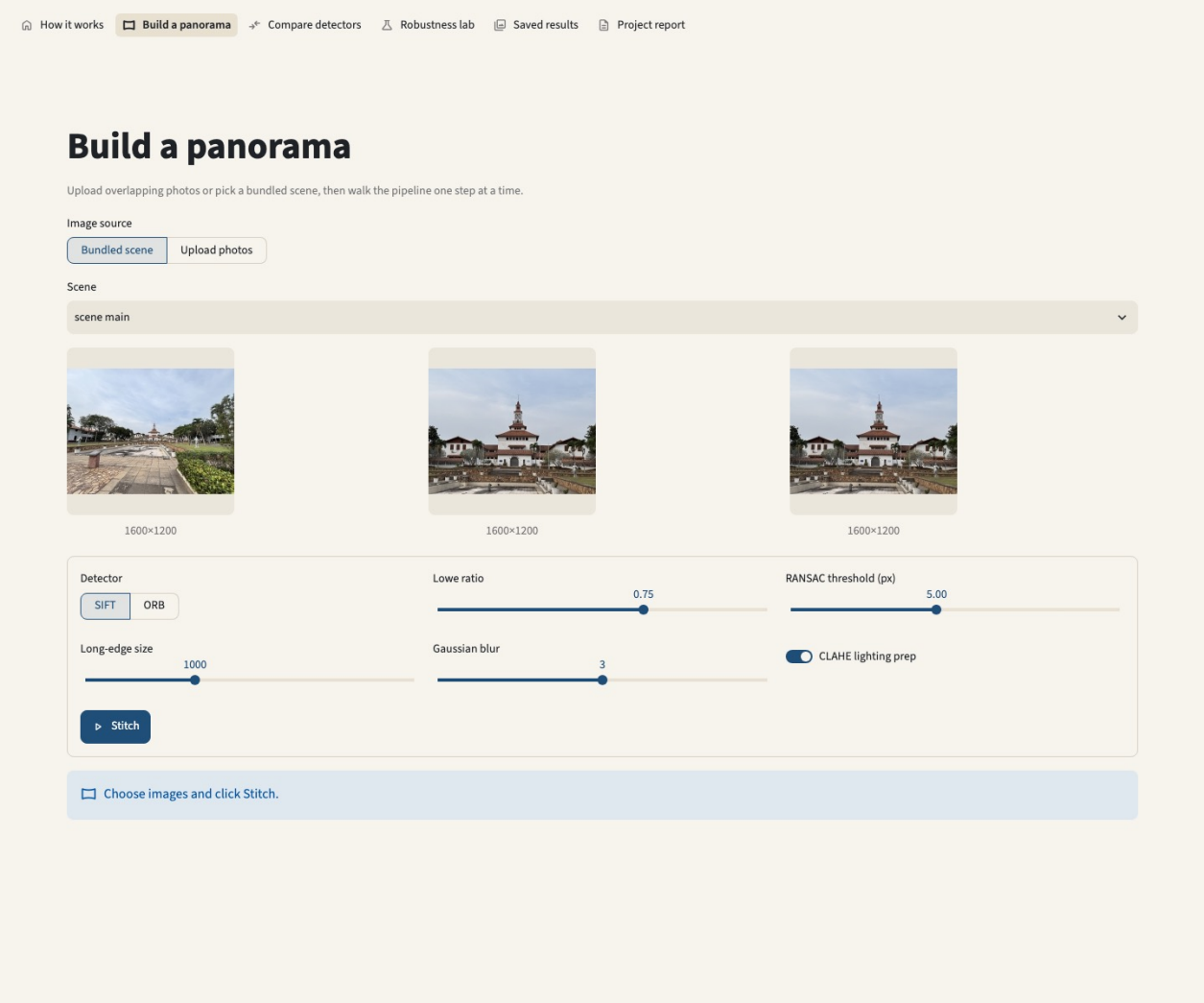


Figure 2: Build a panorama: bundled Balme Library views, SIFT, Lowe ratio 0.75, RANSAC 5 px.

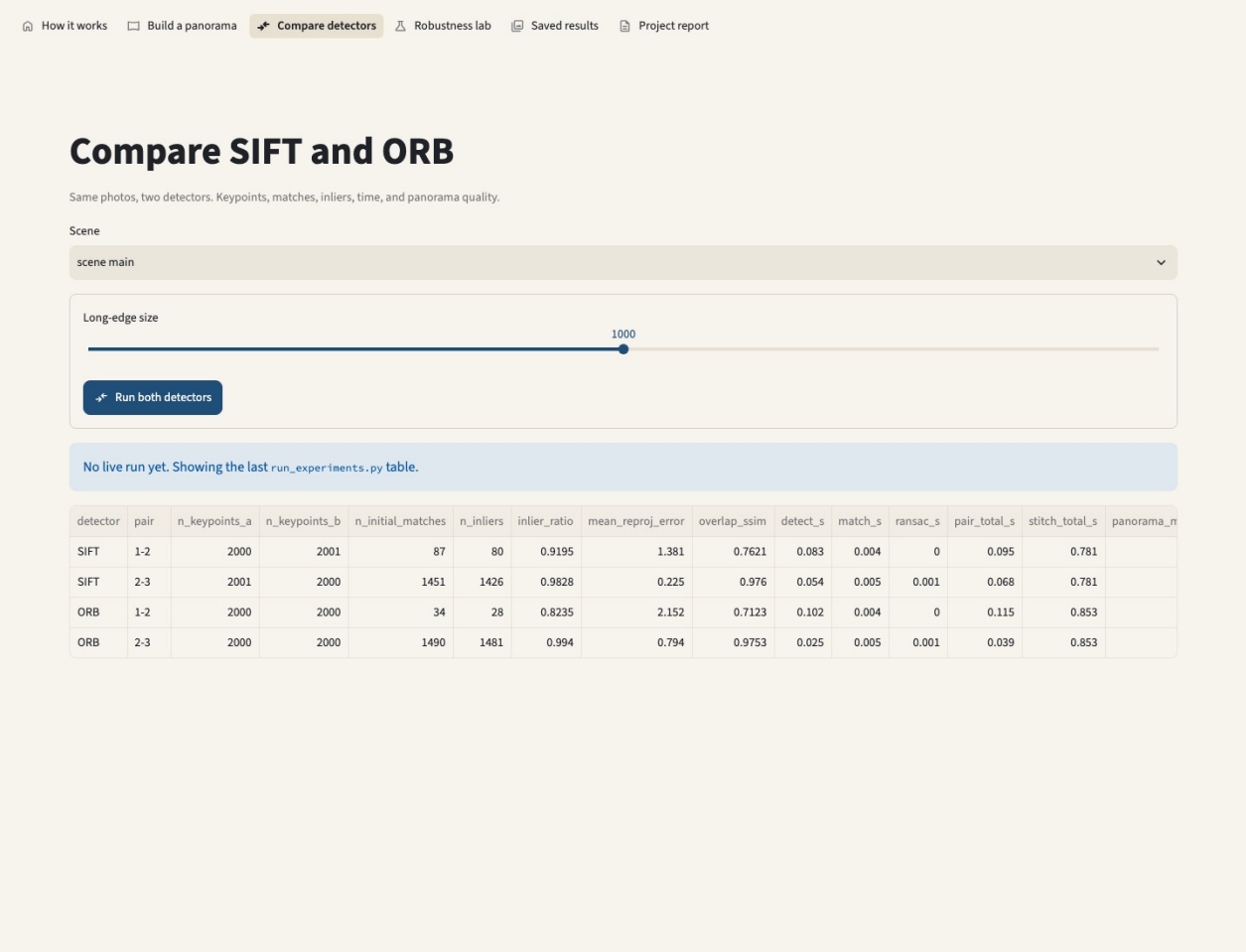


Figure 3: Compare detectors: side-by-side SIFT vs ORB table on the same pair.

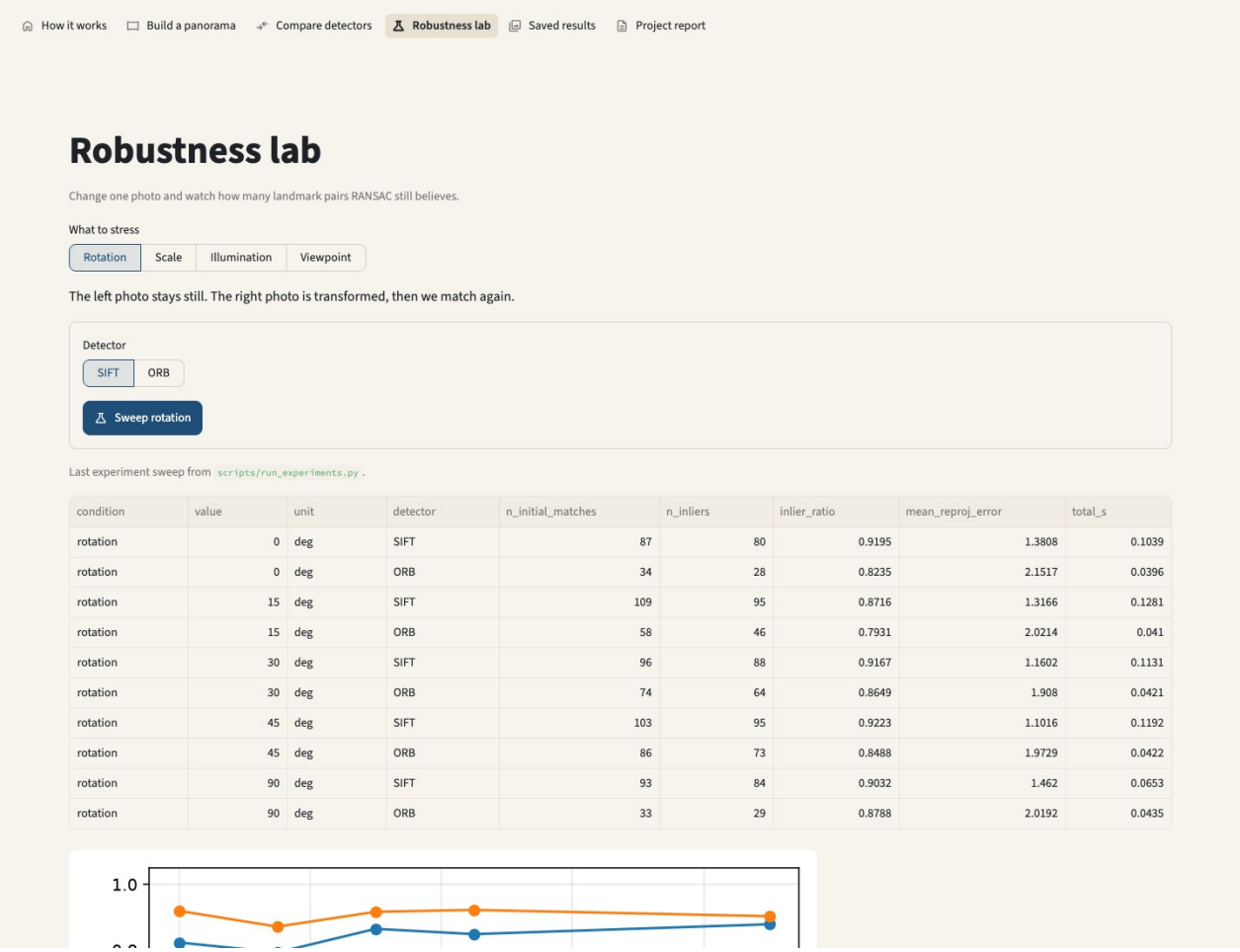


Figure 4: Robustness lab: rotation, scale, and illumination sweeps on a chosen pair.

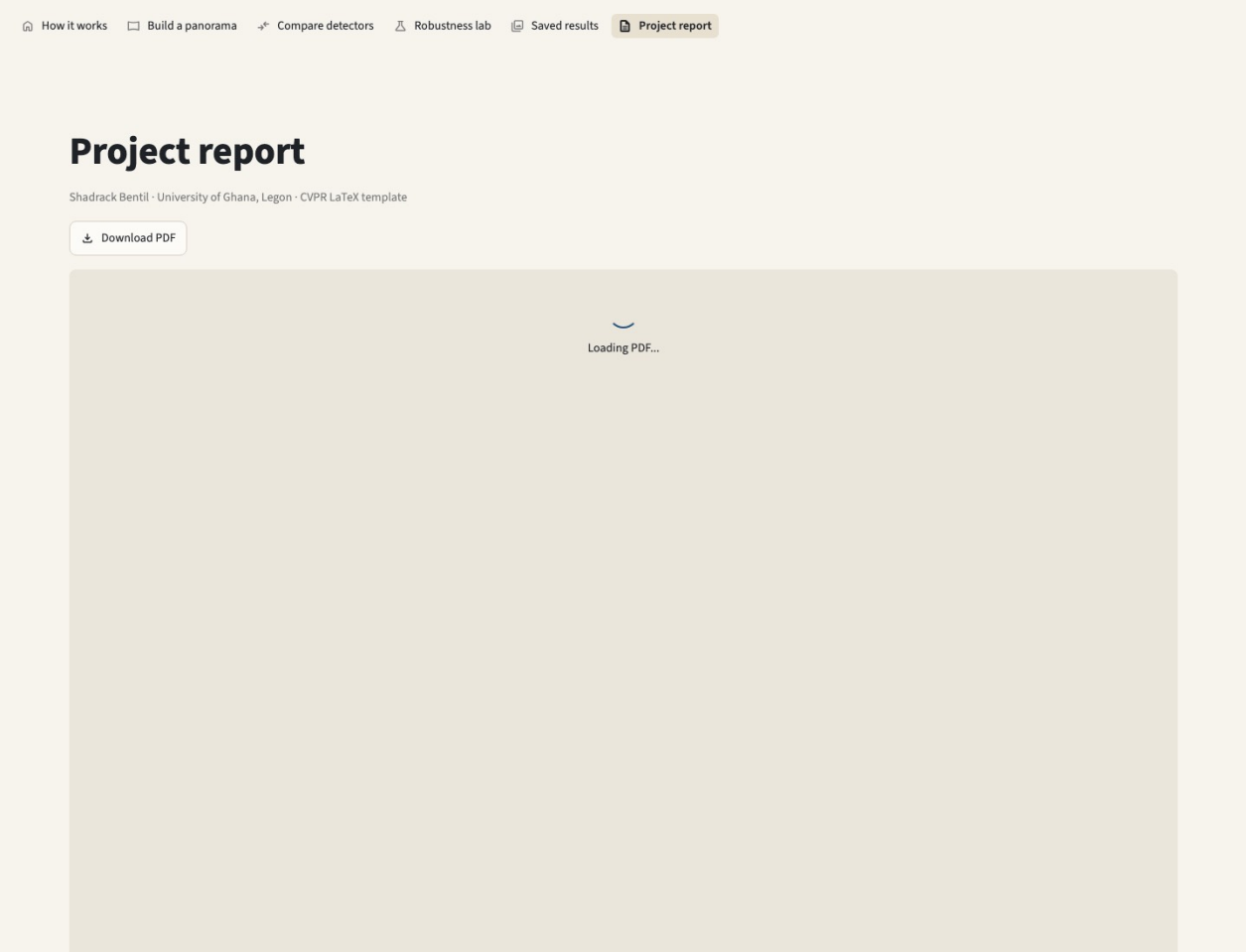


Figure 5: Project report tab: download and embed of the separate CVPR PDF (feature_based_panorama.pdf), not this packet.

5. Results

Primary scene: three Balme Library photographs (Wikimedia Commons, CC BY 4.0). Pair 2–3 is a near-duplicate facade; pair 1–2 is courtyard-to-facade with less overlap. Mean inlier ratio across the two consecutive pairs: SIFT 0.951, ORB 0.909. Full stitch about 0.78–0.85 s. Brightness gain 0.35 leaves SIFT with zero inliers. Dance Department viewpoint frames barely overlap.

Figure 6: Input views for the primary stitch (Balme Library, left to right).

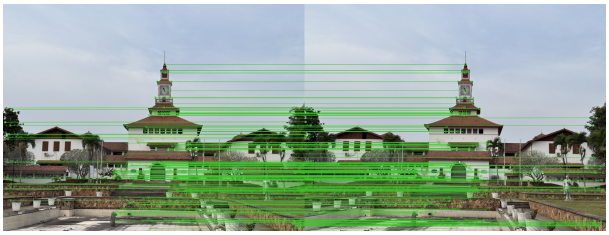
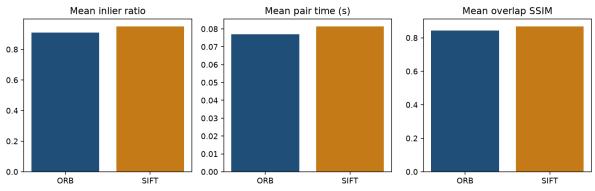


Figure 7: Left: SIFT inliers after RANSAC on a consecutive pair. Right: stitched panorama in the middle-image frame.



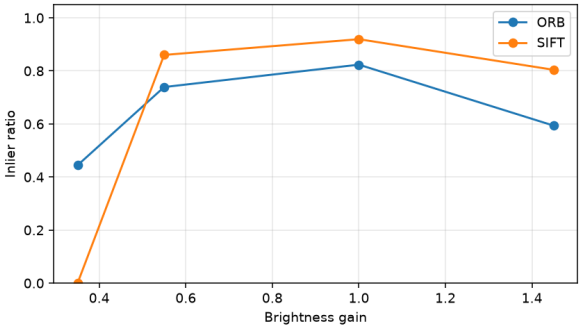


Figure 8: Left: SIFT vs ORB summary on Balme Library. Right: illumination sweep (inlier ratio vs brightness gain).

Table 1: Detector comparison on Balme Library consecutive pairs.

Detector	Pair	Initial	Inliers	Ratio	SSIM
SIFT	1–2	87	80	0.920	0.762
SIFT	2–3	1451	1426	0.983	0.976
ORB	1–2	34	28	0.824	0.712
ORB	2–3	1490	1481	0.994	0.975

SIFT is the tighter geometric estimator on the harder pair (reprojection 1.38 px vs 2.15 px for ORB on pair 1–2). ORB remains competitive when the facade is almost duplicated (pair 2–3). Both methods need genuine overlap: the Dance Department viewpoint set is a negative control and does not produce a usable panorama.

6. How to reproduce

```
pip install -r requirements.txt
python scripts/run_experiments.py
streamlit run streamlit_app.py
```

Or open the live URL in Section 1. Bundled photographs can be replaced with phone captures (about 30–50% overlap, captured left to right) in data/scenes/. Experiment CSVs land in report/results/; figures land in report/figures/. The CVPR paper is rebuilt with python scripts/build_report.py; that command also rebuilds this packet from report/submission.tex.